Version 7.4.7

PVsyst - Simulation report

Grid-Connected System

Project: SPV5_Kolkata_5MWp

Variant: Bifacial 20deg, pitch 7m, 3D sheds, soiling 3pct, 26x565W x 340, 16x SG250HX

Sheds, single array

System power: 4995 kWp

Kolkata - India

Author

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Project summary

Geographical Site
Kolkata
India

Situation

Latitude	22.56 °N
Longitude	88.36 °E
Altitude	11 m
Time zone	UTC+5.5

Project settings

Albedo	0.20

Weather data
Kolkata
Meteonorm 8.1 (1991-2012) - Synthetic

System summary

Grid-Connected System

Sheds, single array

PV Field Orientation
Fixed plane
Tilt/Azimuth 20 / 0 °

Near Shadings
Linear shadings : Fast (table)

User's needs
Unlimited load (grid)

System information

PV Array

Nb. of modules	8840 units
Pnom total	4995 kWp

Inverters

Nb. of units	16 units
Pnom total	4000 kWac
Pnom ratio	1.249

Results summary

Produced Energy	7006164 kWh/year	Specific production	1403 kWh/kWp/year	Perf. Ratio PR	84.78 %

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General parameters

Grid-Connected System — **Sheds, single array**

PV Field Orientation

Orientation	
Fixed plane	
Tilt/Azimuth	20 / 0 °

Sheds configuration		
Nb. of sheds	17	units
Single array		
Sizes		
Sheds spacing	7.00	m
Collector width	4.58	m
Ground Cov. Ratio (GCR)	65.4	%
Top inactive band	0.02	m
Bottom inactive band	0.02	m
Shading limit angle		
Limit profile angle	30.4	°

Models used	
Transposition	Perez
Diffuse	Perez, Meteonorm
Circumsolar	separate

Horizon
Free Horizon

Near Shadings
Linear shadings : Fast (table)

User's needs
Unlimited load (grid)

Bifacial system
Model: 2D Calculation unlimited sheds

Bifacial model geometry		
Sheds spacing	7.00	m
Sheds width	4.62	m
Limit profile angle	30.4	°
GCR	65.9	%
Height above ground	1.00	m

Bifacial model definitions	
Ground albedo	0.25
Bifaciality factor	80 %
Rear shading factor	5.0 %
Rear mismatch loss	10.0 %
Shed transparent fraction	0.0 %

PV Array Characteristics

PV module

Manufacturer	Waaree Energies
Model	BiN-08-565
(Custom parameters definition)	
Unit Nom. Power	565 Wp
Number of PV modules	8840 units
Nominal (STC)	4995 kWp
Modules	340 string x 26 In series
At operating cond. (50°C)	
Pmpp	4620 kWp
U mpp	1034 V
I mpp	4467 A

Inverter

Manufacturer	Sungrow
Model	SG250-HX
(Original PVsyst database)	
Unit Nom. Power	250 kWac
Number of inverters	192 * MPPT 8% 16 units
Total power	4000 kWac
Operating voltage	500-1450 V
Pnom ratio (DC:AC)	1.25
No power sharing between MPPTs	

Total PV power

Nominal (STC)	4995 kWp
Total	8840 modules
Module area	22836 m²

Total inverter power

Total power	4000 kWac
Number of inverters	16 units
Pnom ratio	1.25

Array losses

Array Soiling Losses
Loss Fraction 3.0 %

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 29.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 3.8 mΩ
Loss Fraction 1.5 % at STC

LID - Light Induced Degradation
Loss Fraction 0.5 %

Module Quality Loss
Loss Fraction 0.0 %

Module mismatch losses
Loss Fraction 2.0 % at MPP

Strings Mismatch loss
Loss Fraction 0.1 %

IAM loss factor
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

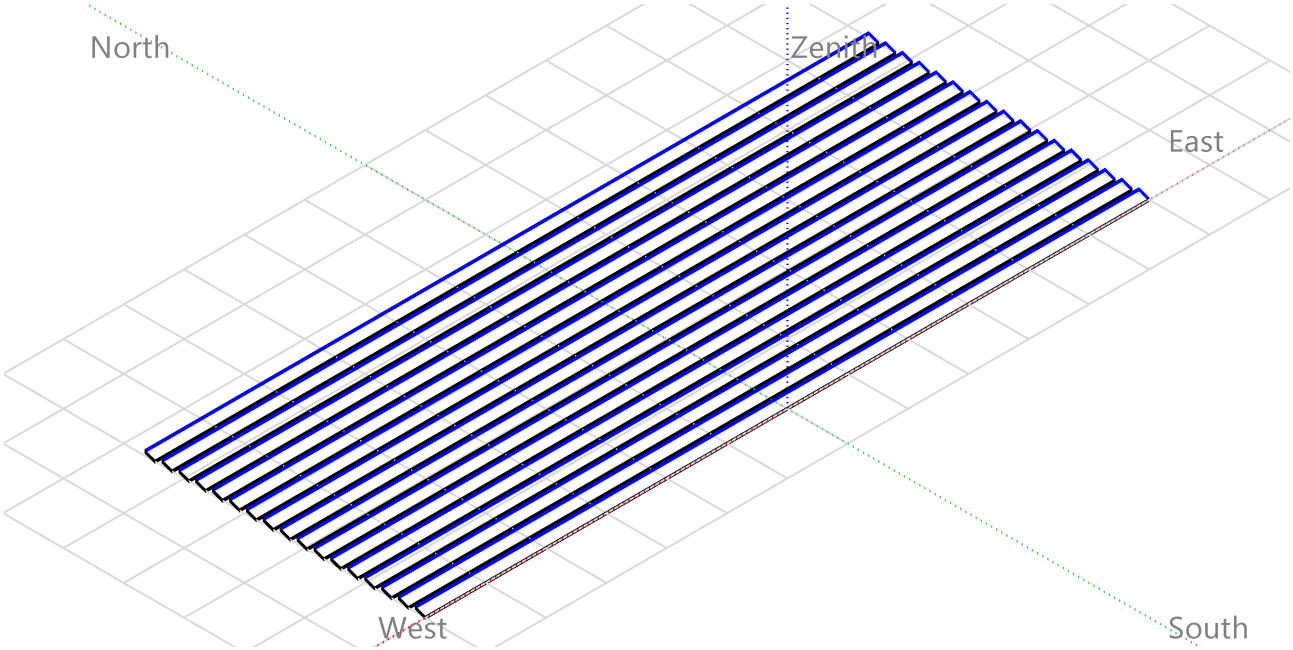
0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000

System losses

Unavailability of the system
Time fraction 1.0 %
3.7 days,
3 periods

Near shadings parameter

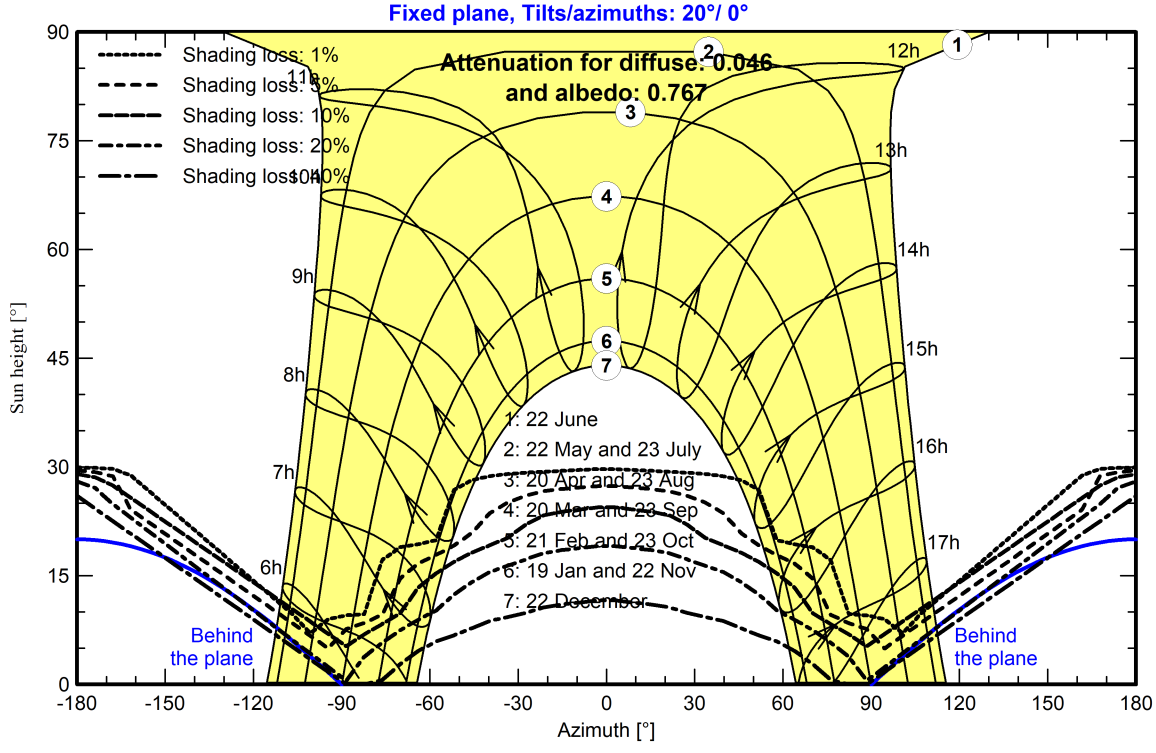
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1

Fixed plane, Tilts/azimuths: 20° / 0°

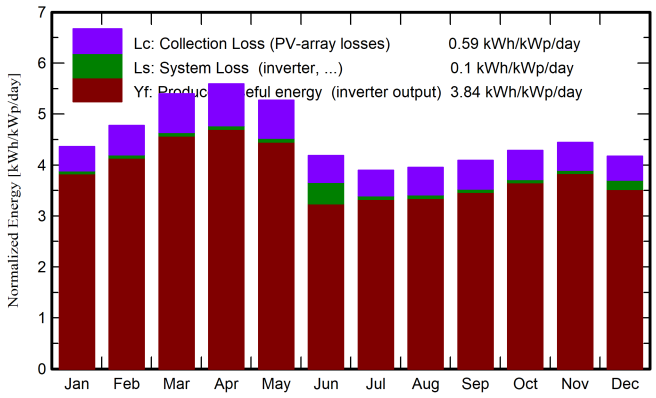


Main results

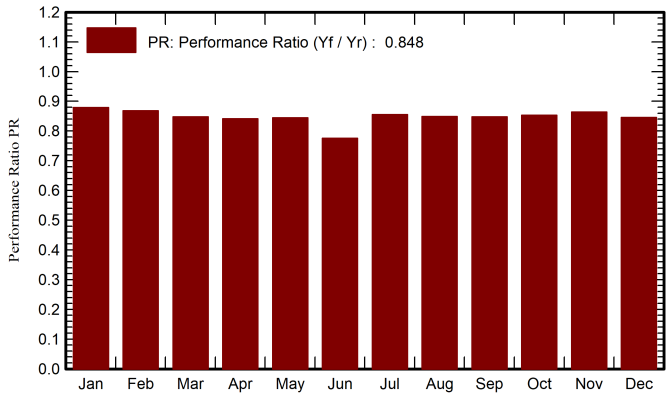
System Production

Produced Energy	7006164 kWh/year	Specific production	1403 kWh/kWp/year
		Perf. Ratio PR	84.78 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

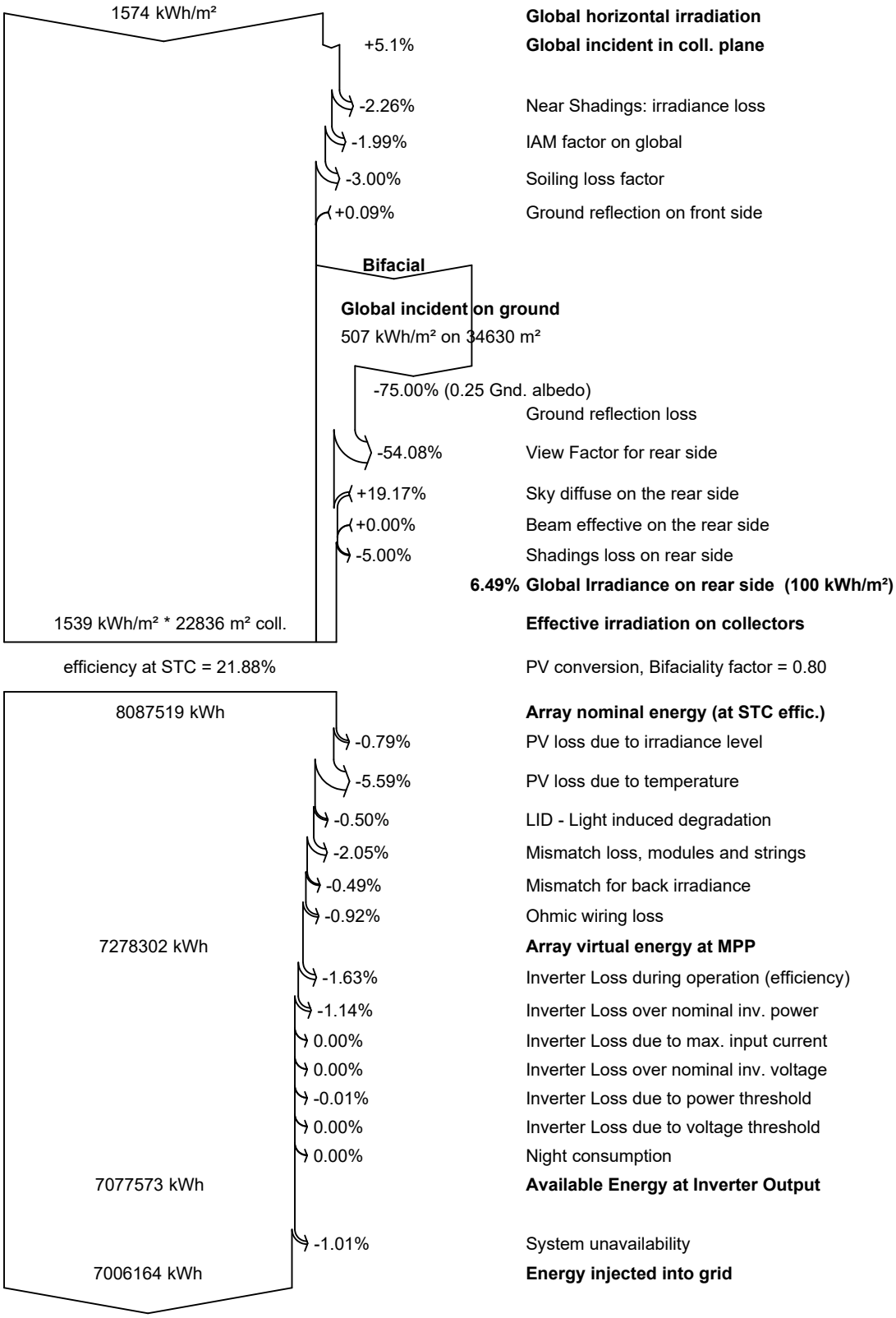
	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	112.1	60.55	17.45	135.2	125.8	602780	593848	0.879
February	118.0	67.00	22.10	133.7	124.9	588327	579599	0.868
March	156.9	85.04	27.55	167.3	156.6	718680	708203	0.847
April	166.5	88.89	30.18	167.7	156.8	715662	705011	0.842
May	171.7	96.70	31.43	163.5	152.2	701253	689886	0.845
June	134.0	93.63	30.47	125.5	115.7	549382	486442	0.776
July	128.3	87.28	29.78	120.8	111.2	526927	516207	0.855
August	125.0	83.58	29.36	122.5	113.2	529310	518895	0.848
September	120.4	78.22	28.31	122.8	113.7	529585	520134	0.848
October	121.8	73.59	27.25	132.9	123.8	576196	566709	0.854
November	112.8	60.86	23.36	133.4	124.5	584473	575463	0.864
December	106.1	57.07	18.68	129.3	120.4	573883	545769	0.845
Year	1573.6	932.42	26.34	1654.6	1538.8	7196457	7006164	0.848

Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

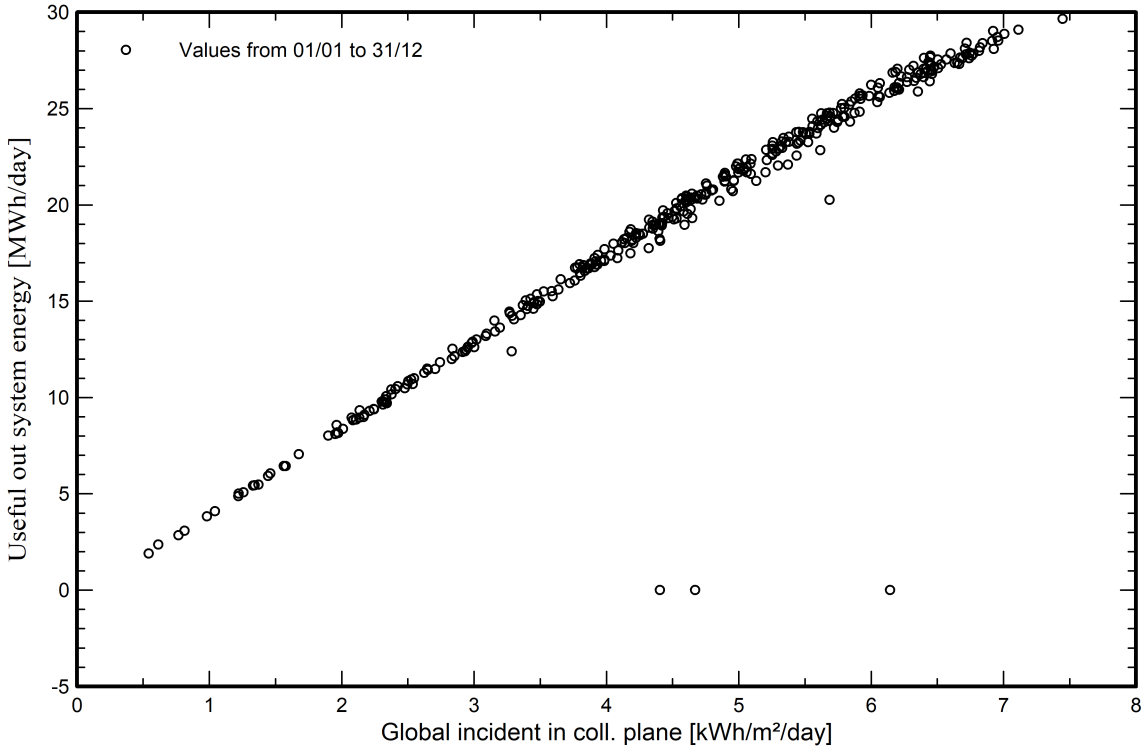
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Loss diagram



Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

